

GTA: Towards Generative Text-To-Audio Retrieval via Multi-Scale Tokenizer

Minghui Fang¹, Shengpeng Ji¹, Jialong Zuo¹, Xize Cheng¹, Wenrui Liu¹, Xiaoda Yang¹, Ruofan Hu¹,
Jieming Zhu², Zhou Zhao^{1,**}

¹Zhejiang University, China

²Huawei Noah's Ark Lab, China

minghuifang@zju.edu.cn, zhaozhou@zju.edu.cn

Abstract

Text-to-audio retrieval is a fundamental task in acoustic signal processing. Currently, mainstream approaches primarily employ a dual-tower architecture, independently encoding text and audio while performing similarity score matching. However, these methods struggle to maintain a uniform embedding space and the latency of score matching increases with corpus size. In this light, we propose GTA to move towards a generative text-to-audio retrieval paradigm. Specifically, we utilize a **multi-scale audio tokenizer** to embed audio semantics into identifiers, while incorporating a **dual-alignment strategy** to ensure consistency between audio and text semantics. Furthermore, we implement curriculum learning to bridge the gap between training and inference, guiding the model toward precise token-level generation. Extensive experiments demonstrate the effectiveness and robustness of GTA, further validating the feasibility of the generative text-to-audio retrieval paradigm.

Index Terms: text-to-audio retrieval, generative retrieval

1. Introduction

Text-to-audio retrieval is a fundamental task in acoustic signal processing, enabling users to intuitively retrieve corresponding audio using natural language descriptions. Current classical methods [1, 2, 3, 4] treat this task as a discriminative learning problem, employing a dual-tower framework to project text queries and candidate audio into a unified high-dimensional embedding space, where audio is ranked based on similarity scores. However, these methods struggle to effectively align data from different modalities [5] and become inefficient when dealing with large-scale corpora. Specifically, due to the significant disparity in semantic representations across modalities, the dual-tower framework typically requires large-scale pretraining to bridge this gap. Additionally, as the number of candidate audios increases, the similarity computation and ranking process introduces higher latency [6].

As an emerging paradigm in document retrieval, generative retrieval [7, 8, 9, 10] introduces a new perspective and possibilities for cross-modal retrieval. As shown in Figure 1 (b), the core idea is to assign a semantic identifier to each candidate target, and then use a language model to replace the traditional retrieval process with the task of generating these identifiers. Compared to traditional retrieval frameworks, generative retrieval searches for the best candidate targets based solely on the user's query, eliminating the need for a unified embedding space and explicit similarity matching. Given the impressive performance of language models [11, 12, 13], it is worthwhile to explore how to effectively integrate them with text-to-audio retrieval. Firstly,

**indicates the corresponding author.

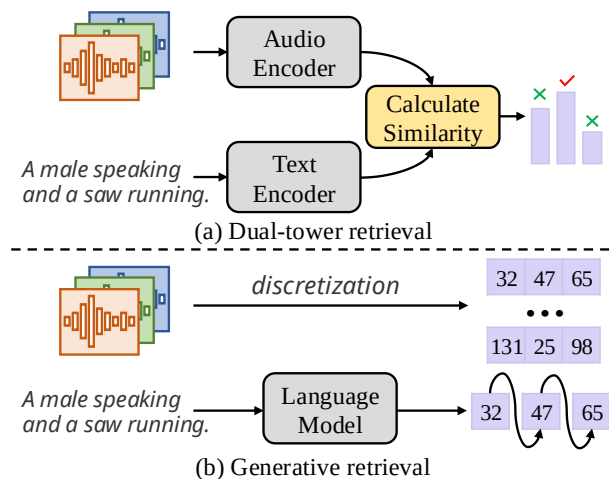


Figure 1: The comparison between traditional dual-tower retrieval and generative retrieval paradigm.

unlike documents that can directly use titles or key sentences as identifiers [14, 15, 16], unstructured audio can not explicitly extract identifiers. Secondly, identifiers serve as a bridge between queries and audio in generative text-to-audio retrieval. However, due to the temporal nature of audio, it often contains redundant content unrelated to the query. Thus, an effective mechanism is needed to align the semantics of the identifiers with the query. Thirdly, unlike text generation tasks, generative retrieval has stricter constraints. In other words, each audio corresponds to an identifier with no interchangeability, which requires that each token be generated precisely.

In this paper, we introduce GTA, which leverages discretization and the power of language models to enable **Generative Text-to-Audio** retrieval. To address the aforementioned challenges, GTA is designed with three key components to support end-to-end text-to-audio retrieval. Specifically, we propose to employ a **multi-scale residual quantizer** as an audio tokenizer to extract semantic information from audio embeddings, taking the closest codeword in each codebook as the identifier. To facilitate autoregressive modeling and match the hierarchical refinement nature of the residuals, the size of the deep codebook is larger than that of the shallow codebook. Meanwhile, to align with the high-level semantics of the text, we design a **dual alignment strategy** to bring the paired identifiers closer to the text. Finally, we employ curriculum learning to enhance the generative process of the language model. By randomly merging ground truth tokens with predicted tokens, we reduce the gap between training and inference.

We conducted extensive experiments to demonstrate the superior performance of GTA, alongside synchronized ablation studies to validate the effectiveness of each component. Additionally, GTA shows stable retrieval efficiency on both CPU and GPU, indicating that it is not affected by the number of candidates. GTA challenges traditional dual-tower retrieval frameworks and provides new insights for text-to-audio retrieval.

Our contributions are highlighted as follows.

- We propose a multi-scale and multi-objective optimized audio tokenizer, which efficiently extracts audio identifiers that are consistent with the textual semantics.
- We reformulated text-to-audio retrieval as an autoregressive generative task and developed a curriculum learning strategy to enhance its performance.
- Experiments on two widely used benchmarks demonstrate the superior retrieval performance and efficiency of GTA. Ablation studies and in-depth analysis confirm the effectiveness and robustness of the individual components.

2. Method

2.1. Overview

GTA is an end-to-end text-to-audio retrieval framework, which takes natural language queries as input and generates corresponding audio identifiers. This pipeline eliminates the need for auditory input, requiring only the generation training of {Query, Identifier} pairs. The whole framework is divided into two modules: 1) Multi-scale audio tokenizer, which is used to construct semantic identifiers for audio, and 2) Curriculum-learning based generative training, which utilizes a language model to convert discriminative retrieval to generative retrieval.

2.2. Multi-Scale Audio Tokenizer

The current popular approach [17, 18, 19, 20, 21] for unstructured audio data is to discretize it, making it compatible with autoregressive generation models. Essentially, these methods are based on vector quantization [22, 23], embedding the semantic information of audio into codebooks and then selecting the most matching codeword from multiple codebooks. However, they tend to compress at the ratio of hundreds of tokens per second and retain too much acoustic information, which focuses on audio reconstruction and high fidelity. As previously mentioned, identifiers are not interchangeable for the retrieval task, restricting the autoregressive model to accurately generate hundreds of tokens is extremely difficult. Meanwhile, the retrieval focuses on semantic relevance to the query rather than acoustic features. These drive us to design an audio tokenizer tailored specifically for retrieval tasks.

The audio tokenizer is shown in Figure 2. We encode the paired audio a and text t into embeddings $\hat{a}$ and $\hat{t}$ with the pre-trained CLAP [1]. The formula is as follows:

$$\begin{aligned}\hat{a} &= \text{CLAP}_a(a), \quad \hat{a} \in \mathbb{R}^{1 \times 1024} \\ \hat{t} &= \text{CLAP}_t(t), \quad \hat{t} \in \mathbb{R}^{1 \times 1024}\end{aligned}\quad (1)$$

To further refine the semantics and alleviate the burden of codebook learning, we further downsample with a shallow multilayer perceptron (MLP). Inspired by SpeechT5 [24], the MLP is shared between audio and text to facilitate the initial extraction of their consistent semantics. The formula is as follows:

$$\begin{aligned}\bar{a} &= \text{Encoder}(\hat{a}), \quad \bar{a} \in \mathbb{R}^{1 \times 64} \\ \bar{t} &= \text{Encoder}(\hat{t}), \quad \bar{t} \in \mathbb{R}^{1 \times 64}\end{aligned}\quad (2)$$

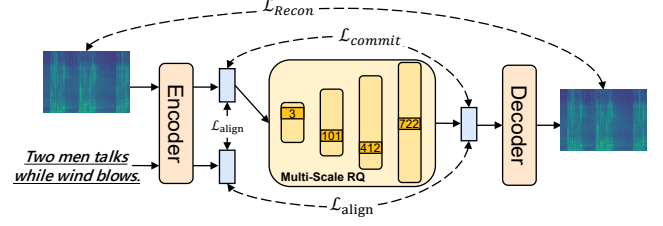


Figure 2: The pipeline of multi-scale audio tokenizer.

After that we perform residual quantization, where the quantizer $Q = \{C_1, C_2, \dots, C_N\}$ has N codebooks and the codebook $C_n = \{v_n^1, v_n^2, \dots, v_n^M\}$, where M denotes the total number of codewords in the codebook, and v_n^m denotes the m -th codeword in the n -th codebook. The initial residual $r_0 = \bar{a}$, and then search for the closest codeword v_1^k in the codebook C_1 . The index k of the codeword v_1^k is considered as the identifier token obtained from the codebook C_1 . Following the suggestion of prior work [25], we use cosine similarity as the distance function to improve code usage with the following formula:

$$\begin{cases} \cos(r_0, v_n^m) = \frac{r_0 \cdot v_n^m}{\|r_0\| \|v_n^m\|} \\ k_1 = \arg \max_{k \in \{1, 2, \dots, M\}} \cos(r_0, v_1^k) \end{cases} \quad (3)$$

The residuals are then updated to $r_1 = r_0 - v_1^{k_1}$, this process is executed recursively N times and the indexes $\{k_1, k_2, \dots, k_N\}$ extracted from all codebooks are considered as identifiers of the audio. Finally, $Q(\bar{a}) = \sum_{i=1}^N v_i^{k_i}$ is passed to the decoder, whose structure is symmetrical to the encoder. We treat the stepwise construction of identifiers as a gradual partitioning of categories. Meanwhile, considering the nature of autoregressive modeling, the difficulty of predicting tokens should increase from easy to hard, as incorrect predictions of earlier tokens render subsequent predictions meaningless. Based on these considerations, we designed multi-scale codebooks, where the size of the codebook gradually increases from the shallow to the deep layers.

In order to efficiently train the audio tokenizer, we retain the classical reconstruction loss and commit loss, where

$$\begin{aligned}\mathcal{L}_{recon} &= \|\hat{a} - \text{Decoder}(Q(\text{Encoder}(\hat{a})))\|_2^2, \\ \mathcal{L}_{commit} &= \sum_{i=1}^N \|sg[r_i] - v_i^{k_i}\|_2^2 + \|r_i - sg[v_i^{k_i}]\|_2^2\end{aligned}\quad (4)$$

Here, sg is the stop-gradient operation [26]. The reconstruction loss ensures that the identifiers capture the essential audio semantics, while the commit loss prevents excessive deviation between the encoder output and the quantized vectors, thereby avoiding codebook collapse.

To further bridge the gap in semantics between queries and audio, we propose a dual alignment strategy to further enhance the tokenizer. As mentioned earlier, we aim to extract consistent high-level semantics between text and audio using the shared MLP. To achieve this goal, we introduce an optimization objective here that we hope to align with the query before quantization. Correspondingly, we introduce another optimization objective after quantization which preserves semantics by limiting the distance between the codebook and the query. The formula is as follows:

$$\mathcal{L}_{dual.align} = \|\bar{t} - \bar{a}\|_2^2 + \|\bar{t} - z\|_2^2 \quad (5)$$

where z denotes the quantized audio latent states. The total training loss of the audio tokenizer can be defined as:

$$\mathcal{L}_{tokenizer} = \mathcal{L}_{recon} + \mathcal{L}_{commit} + \mathcal{L}_{dual.align} \quad (6)$$

2.3. Curriculum Learning-Based Generative Retrieval

With the help of the audio tokenizer, the paired {Query, Audio} is converted into {Query, Identifier}, at which point the retrieval task can be viewed as end-to-end text generation. Since tokens come from different codebooks, the same number in different positions carries different semantics. For example, 3_1 and 3_2 in $(3_1, 2, 3_2, 1)$ represent different meanings. Considering this, we add special prefixes to each position to distinguish them, e.g. (a_3, b_2, c_3, d_1) , and add them to the language model vocabulary. Based on the encoder-decoder architecture, the cross-entropy loss for generating semantic identifiers T given a query q can be written as:

$$\mathcal{L}_{\text{generation}} = - \sum_{i=1}^L \log p(t_i | E(q), t_{<i}, \theta_i) \quad (7)$$

where t_i is the i -th token, L denotes the length of T and $E(\cdot)$ denotes the encoder of language models, θ denotes the total parameters and θ_i is the parameter for the i -th step.

Given that each audio corresponds to only one identifier, this requires token generation to be sufficiently precise. Considering that during the training stage, the model can always refer to the groundtruth to generate the next token, while in the inference stage, the model generates the next token based solely on its previous predictions, we are motivated to bridge this gap. Therefore, we introduce curriculum learning [27] to assist the model. Specifically, the model performs the first forward propagation to obtain logits, and we decide with probability P whether to use groundtruth t_i or model predictions $\hat{t}_i$ for each token, which are mixed and passed to the decoder again. The formula is as follows,

$$t'_i = \begin{cases} t_i & \text{with probability } P \\ \hat{t}_i & \text{with probability } 1 - P \end{cases} \quad (8)$$

To obtain the ranked list of audio, we use beam search to generate the top-k audio with the highest relevance to the given query. Meanwhile, we use constrained generation [8, 28] in the decoder network to ensure generating valid identifiers.

3. Experiments

3.1. Datasets and Baselines

We evaluate GTA on two widely used datasets: Audiocaps [29] and Clotho [30]. Audiocaps consist of 39,597 audio samples where each audio sample in the validation and test sets has five captions, making a total of 45,513 {Query, Audio} pairs. Clotho consists of 4981 audio samples with five captions per audio sample, for a total of 24,905 {Query, Audio} pairs.

We select baselines based on the classical dual-tower architecture. The CLAP series [3, 31, 4] explores different combinations of text encoders and audio encoders, presenting an efficient way to handle variable-length audio. While VALOR [32], ONE-PIECE [33] aim to build omni-modality fundamental models with larger number of parameters and pre-training scale, which consider text-to-audio retrieval as a subtask. Consistent with previous work [2, 3], we use the standard retrieval metric $R@K$ where K is set to 1, 5, 10.

3.2. Implementation Details

Following the previous work [8, 10, 34], we generate five additional captions for each audio as Query for data enhancement.

Table 1: *Performance comparisons on text-to-audio retrieval tasks, all results from the original paper.*

Baseline	AudioCaps			Clotho		
	R@1	R@5	R@10	R@1	R@5	R@10
LAION-CLAP [3]	36.1	71.8	83.9	16.1	38.3	51.1
CNN14-BERT [31]	35.1	70.0	82.1	21.5	47.9	61.9
HTSAT-BERT [31]	42.2	76.5	87.1	19.7	45.7	59.4
CLAP-HTSAT [4]	34.6	70.2	82	16.7	41.1	54.1
VALOR-B [32]	40.1	73.9	83.1	17.5	42.7	55.3
ONE-PIECE [33]	42.5	77.5	88.4	22.4	49.0	62.7
GTA(Ours)	48.5	78.0	83.6	44.1	66.3	74.8

The audio tokenizer is trained with learning rate of 0.00001 and batch size of 512. The encoder consists of 4 intermediate layers with sizes of 768, 512, 256, and 128, respectively, and the decoder is completely symmetrical to it. The quantizer includes four codebooks with sizes of 128, 256, 512 and 1024, each codebook vector has a dimension of 64. Following [35] suggestion, we are initializing the codebook centroid using K-Means to ensure training stability. The generative model utilizes T5-base [11] as the backbone and has an initial learning rate of $1e-7$ during the warm up stage, after which training is completed with a learning rate of 0.001 and a batch size of 128. The probability P of curriculum learning has an initial value of 1 and decreases by 0.005 per epoch.

3.3. Main Results

In Table 1, we report the the performance of the baseline and GTA. We observe that all methods perform better on AudioCaps than on Clotho. Compared to the baseline, GTA performs competitively on AudioCaps and outperforms the baseline across the board on Clotho. Our analysis indicates that the audio duration in the Clotho ranges from 15 to 30 seconds, whereas the audio in AudioCaps are generally around 10 seconds. Enhancing the retrieval performance for long and variable-length audio sequences is exactly the challenge [3] that the dual-tower model aims to address. In contrast, the audio semantics are already embedded in the parameters of the language model during training. GTA only needs to focus on processing text queries and predicting fixed-length audio identifiers, which alleviates the retrieval burden to some extent.

3.4. Ablation Results

We employ multi-scale codebooks in the audio tokenizer to match the progressively refined nature of residual transmission. Additionally, we use dual-alignment to ensure that the identifiers not only encapsulate audio semantics but also align with the text. During the training stage, we enhance the language model’s prediction of identifiers through curriculum learning. To validate the effectiveness of these key components, we report the results of the ablation study in Table 2.

We set the size of all codebooks to 128 to simulate the w/o multi-scale setting, and the retrieval performance showed significant degradation. Intuitively, we believe that as the residuals are passed from level to level, deeper codebook captures finer semantic granularity. With increasing depth of the codebook, extending the size of the codebook facilitates improved quantization accuracy. Removing dual-alignment during the audio tokenizer training also heavily impacted retrieval performance, highlighting the importance of aligning identifiers with query semantics. Finally, removing the curriculum learning reduced retrieval performance, indicating that assisting the

Table 2: Ablation studies on key components of GTA.

Setting	R@1	R@5	R@10
w/o multi-scale	38.4	57.3	65.2
w/o dual-alignment	38.5	58.7	66.7
w/o curriculum learning	41.1	64.8	73.8
GTA(Ours)	44.1	66.3	74.8

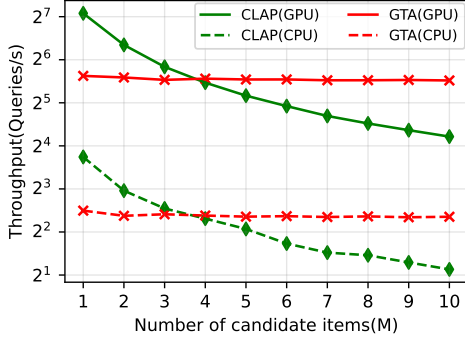


Figure 3: The efficiency of CLAP and GTA are measured by throughput (queries processed per second).

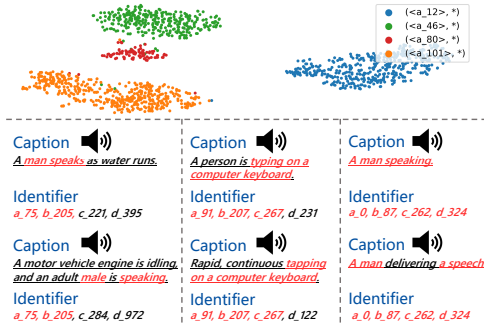


Figure 4: The t-SNE visualization of audio embeddings with the same prefix and the more intuitive case studies.

model’s learning at the token level to bridge the gap between training and inference is meaningful.

3.5. Comparison of Inference Latency

We compare the retrieval efficiency of CLAP [1] and GTA on both CPU and NVIDIA V100 GPU to evaluate the performance of dual-tower and generative frameworks. We simulate 50 concurrent queries and change the number of candidate audios, the metric is throughput (queries/per second) and the detailed results are shown in Figure 3. We pre-encode the audio into embeddings, so the primary time cost of CLAP comes from text encoding, similarity matching, and ranking. With the increase of audio, the retrieval throughput of the dual-tower framework gradually decreases. In contrast, GTA generates identifiers directly based on the query, with efficiency independent of the number of candidate audios, which is an advantage of directly encoding all audios as its parameters.

Table 3: Performance of varying token size.

Token size	R@1	R@5	R@10
[128, 128, 128, 128]	38.4	57.3	65.2
[128, 512, 768, 1024](Ours)	44.1	66.3	74.8
[1024, 1024, 1024, 1024]	35.7	55.1	63.4

Table 4: Performance of varying token length.

Token length	R@1	R@5	R@10
4(Ours)	44.1	66.3	74.8
5	43.9	66.4	75.0
6	43.6	66.0	74.7

3.6. Deep Analysis on Audio Tokenzier

3.6.1. Analysis through visualisation.

We further investigate the effectiveness of the audio tokenizer. As shown in Figure 4, we visualize the correlation between the audio, the corresponding query, and the identifiers. We use t-SNE to visualize the distribution of audio embeddings with the same prefix, showing clear clustering patterns. Embeddings within clusters are more closely grouped, while those between clusters are more dispersed. Additionally, we conducted a more detailed case study. It can be observed that the closer the identifiers are, the more similar the semantics of the corresponding audio queries. This demonstrates that the identifiers extracted by the audio tokenizer not only capture the semantics of the audio but also align with the semantics of the query.

3.6.2. Analysis on token size.

We verify the effectiveness of multi-scale by adjusting the size of the codebooks, with the experimental results on Clotho shown in Table 3. We observe that both excessively large and small codebook sizes degrade retrieval performance, as the size of the codebook controls the granularity of classification, thus limiting the expressive power of the semantic identifiers. Additionally, an overly large decoding space reduces the accuracy of autoregressive model predictions. In contrast, the multi-scale approach effectively alleviates these issues.

3.6.3. Analysis on token length.

We adjust the length of the identifiers by varying the number of codebooks, with the experimental results on Clotho shown in Table 4. We observe only slight fluctuations in retrieval performance, indicating the effectiveness of the semantic identifiers and the stability of the retrieval model. Although increasing the identifier length helps distinguish different candidates at a finer granularity, excessively long identifiers reduce codebook utilization and increase decoding steps. Therefore, we ultimately choose to use four tokens.

4. conclusion

In this paper, we introduce GTA, an innovative end-to-end text-to-audio retrieval framework. By combining a multi-scale audio tokenizer with a dual alignment strategy, GTA effectively constructs audio identifiers that are consistent with text semantics. With the power of generative models and curriculum learning, GTA demonstrates outstanding retrieval performance and efficiency across two benchmark datasets. Ablation studies and in-depth analysis also confirm the robustness of this paradigm.

5. References

- [1] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, “Clap learning audio concepts from natural language supervision,” in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [2] B. Elizalde, S. Deshmukh, and H. Wang, “Natural language supervision for general-purpose audio representations,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 336–340.
- [3] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, “Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation,” in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [4] S. Deshmukh, B. Elizalde, and H. Wang, “Audio retrieval with wavtext5k and clap training,” *arXiv preprint arXiv:2209.14275*, 2022.
- [5] S. Lou, X. Xu, M. Wu, and K. Yu, “Audio-text retrieval in context,” in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 4793–4797.
- [6] G. Geigle, J. Pfeiffer, N. Reimers, I. Vulić, and I. Gurevych, “Retrieve fast, rerank smart: Cooperative and joint approaches for improved cross-modal retrieval,” *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 503–521, 2022.
- [7] Y. Tay, V. Tran, M. Dehghani, J. Ni, D. Bahri, H. Mehta, S. Qin, K. Hui, Z. Zhao, J. Gupta *et al.*, “Transformer memory as a differentiable search index,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 21 831–21 843, 2022.
- [8] Y. Wang, Y. Hou, H. Wang, Z. Miao, S. Wu, Q. Chen, Y. Xia, C. Chi, G. Zhao, Z. Liu *et al.*, “A neural corpus indexer for document retrieval,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 25 600–25 614, 2022.
- [9] S. Rajput, N. Mehta, A. Singh, R. Hulikal Keshavan, T. Vu, L. Heldt, L. Hong, Y. Tay, V. Tran, J. Samost *et al.*, “Recommender systems with generative retrieval,” *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [10] W. Sun, L. Yan, Z. Chen, S. Wang, H. Zhu, P. Ren, Z. Chen, D. Yin, M. Rijke, and Z. Ren, “Learning to tokenize for generative retrieval,” *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [11] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, “Exploring the limits of transfer learning with a unified text-to-text transformer,” *Journal of machine learning research*, vol. 21, no. 140, pp. 1–67, 2020.
- [12] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat *et al.*, “Gpt-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023.
- [13] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan *et al.*, “The llama 3 herd of models,” *arXiv preprint arXiv:2407.21783*, 2024.
- [14] Y. Zhou, J. Yao, Z. Dou, L. Wu, P. Zhang, and J.-R. Wen, “Ultron: An ultimate retriever on corpus with a model-based indexer,” *arXiv preprint arXiv:2208.09257*, 2022.
- [15] M. Bevilacqua, G. Ottaviano, P. Lewis, S. Yih, S. Riedel, and F. Petroni, “Autoregressive search engines: Generating substrings as document identifiers,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 31 668–31 683, 2022.
- [16] Y. Tang, R. Zhang, J. Guo, J. Chen, Z. Zhu, S. Wang, D. Yin, and X. Cheng, “Semantic-enhanced differentiable search index inspired by learning strategies,” in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 4904–4913.
- [17] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, “High fidelity neural audio compression,” *arXiv preprint arXiv:2210.13438*, 2022.
- [18] R. Kumar, P. Seetharaman, A. Luebs, I. Kumar, and K. Kumar, “High-fidelity audio compression with improved rvqgan,” *Advances in Neural Information Processing Systems*, vol. 36, 2024.
- [19] S. Ji, M. Fang, Z. Jiang, R. Huang, J. Zuo, S. Wang, and Z. Zhao, “Language-codec: Reducing the gaps between discrete codec representation and speech language models,” *arXiv preprint arXiv:2402.12208*, 2024.
- [20] S. Ji, Z. Jiang, X. Cheng, Y. Chen, M. Fang, J. Zuo, Q. Yang, R. Li, Z. Zhang, X. Yang *et al.*, “Wavtokenizer: an efficient acoustic discrete codec tokenizer for audio language modeling,” *arXiv preprint arXiv:2408.16532*, 2024.
- [21] M. Fang, S. Ji, J. Zuo, H. Huang, Y. Xia, J. Zhu, X. Cheng, X. Yang, W. Liu, G. Wang *et al.*, “Ace: A generative cross-modal retrieval framework with coarse-to-fine semantic modeling,” *arXiv preprint arXiv:2406.17507*, 2024.
- [22] A. Van Den Oord, O. Vinyals *et al.*, “Neural discrete representation learning,” *Advances in neural information processing systems*, vol. 30, 2017.
- [23] D. Lee, C. Kim, S. Kim, M. Cho, and W.-S. Han, “Autoregressive image generation using residual quantization,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 11 523–11 532.
- [24] J. Ao, R. Wang, L. Zhou, C. Wang, S. Ren, Y. Wu, S. Liu, T. Ko, Q. Li, Y. Zhang *et al.*, “Speecht5: Unified-modal encoder-decoder pre-training for spoken language processing,” *arXiv preprint arXiv:2110.07205*, 2021.
- [25] J. Yu, X. Li, J. Y. Koh, H. Zhang, R. Pang, J. Qin, A. Ku, Y. Xu, J. Baldridge, and Y. Wu, “Vector-quantized image modeling with improved vqgan,” *arXiv preprint arXiv:2110.04627*, 2021.
- [26] A. Van Den Oord, O. Vinyals *et al.*, “Neural discrete representation learning,” *Advances in neural information processing systems*, vol. 30, 2017.
- [27] T. Mihaylova and A. F. Martins, “Scheduled sampling for transformers,” *arXiv preprint arXiv:1906.07651*, 2019.
- [28] N. De Cao, G. Izacard, S. Riedel, and F. Petroni, “Autoregressive entity retrieval,” *arXiv preprint arXiv:2010.00904*, 2020.
- [29] C. D. Kim, B. Kim, H. Lee, and G. Kim, “Audiocaps: Generating captions for audios in the wild,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 119–132.
- [30] K. Drossos, S. Lipping, and T. Virtanen, “Clotho: An audio captioning dataset,” in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 736–740.
- [31] X. Mei, C. Meng, H. Liu, Q. Kong, T. Ko, C. Zhao, M. D. Plumbley, Y. Zou, and W. Wang, “Wavcaps: A chatgpt-assisted weakly-labelled audio captioning dataset for audio-language multimodal research,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2024.
- [32] S. Chen, X. He, L. Guo, X. Zhu, W. Wang, J. Tang, and J. Liu, “Valor: Vision-audio-language omni-perception pretraining model and dataset,” *arXiv preprint arXiv:2304.08345*, 2023.
- [33] P. Wang, S. Wang, J. Lin, S. Bai, X. Zhou, J. Zhou, X. Wang, and C. Zhou, “One-peace: Exploring one general representation model toward unlimited modalities,” *arXiv preprint arXiv:2305.11172*, 2023.
- [34] S. Zhuang, H. Ren, L. Shou, J. Pei, M. Gong, G. Zuccon, and D. Jiang, “Bridging the gap between indexing and retrieval for differentiable search index with query generation,” *arXiv preprint arXiv:2206.10128*, 2022.
- [35] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, “Soundstream: An end-to-end neural audio codec,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 495–507, 2021.